Bench exercise: a splice you lower into place

Japanese Joinery by Hand · Episode 11. Build the okkake daisen-tsugi 追掛け大栓継ぎ in foam board or stiff card at one third scale. You'll see why the upper piece goes in from above, how the step and the end tongues lock it, and what the pegs are for. Craft knife, steel rule, square, two short dowels or skewers. About 60 minutes.

1 · Two blocks

1. Make two long boxes, 40 mm wide and 60 mm deep (one third of the episode's 120 × 180 mm).
2. On each top face, mark a splice 120 mm long from the end (twice the depth, like the model), and a center line.

2 · Cut each piece to half width

1. Over the splice, keep the half of the width on one side of the center line, at full depth. At the middle of the splice, step the line across by 5 mm: that's the ago.
2. At the tip, leave a tongue 5 mm wide and 5 mm long (the mechigai), and cut a matching 5 × 5 mm housing at the shoulder for the other piece's tongue.
3. Make the second piece the same, turned end for end, so the two fit side by side.

3 · Put it together

1. Lay the lower piece down. Lower the upper piece from above, beside it. Can it go in from the side? From the end?
2. Pull along the beam, then push sideways. Then lift: nothing stops it.
3. Pierce two holes through both pieces from the sides, a quarter of the splice length either side of the middle, and push in the pegs from opposite sides. Now try lifting.

What to notice

- The step and the tongues do the locking along and across the beam. The pegs stop it lifting out.
- In wood, the step's face is cut on a slight slope, so lowering the upper piece draws the shoulders tight. Card won't show that; try sanding one step face to a slight slope and feel it wedge.

Proportions follow one lab's test specimens (Toda 2012), not a carpentry standard. Carpenters' rules give the splice about 3× the depth (a training course) or 3–4× (a joinery handbook); one test database suggests 1.5–3×.